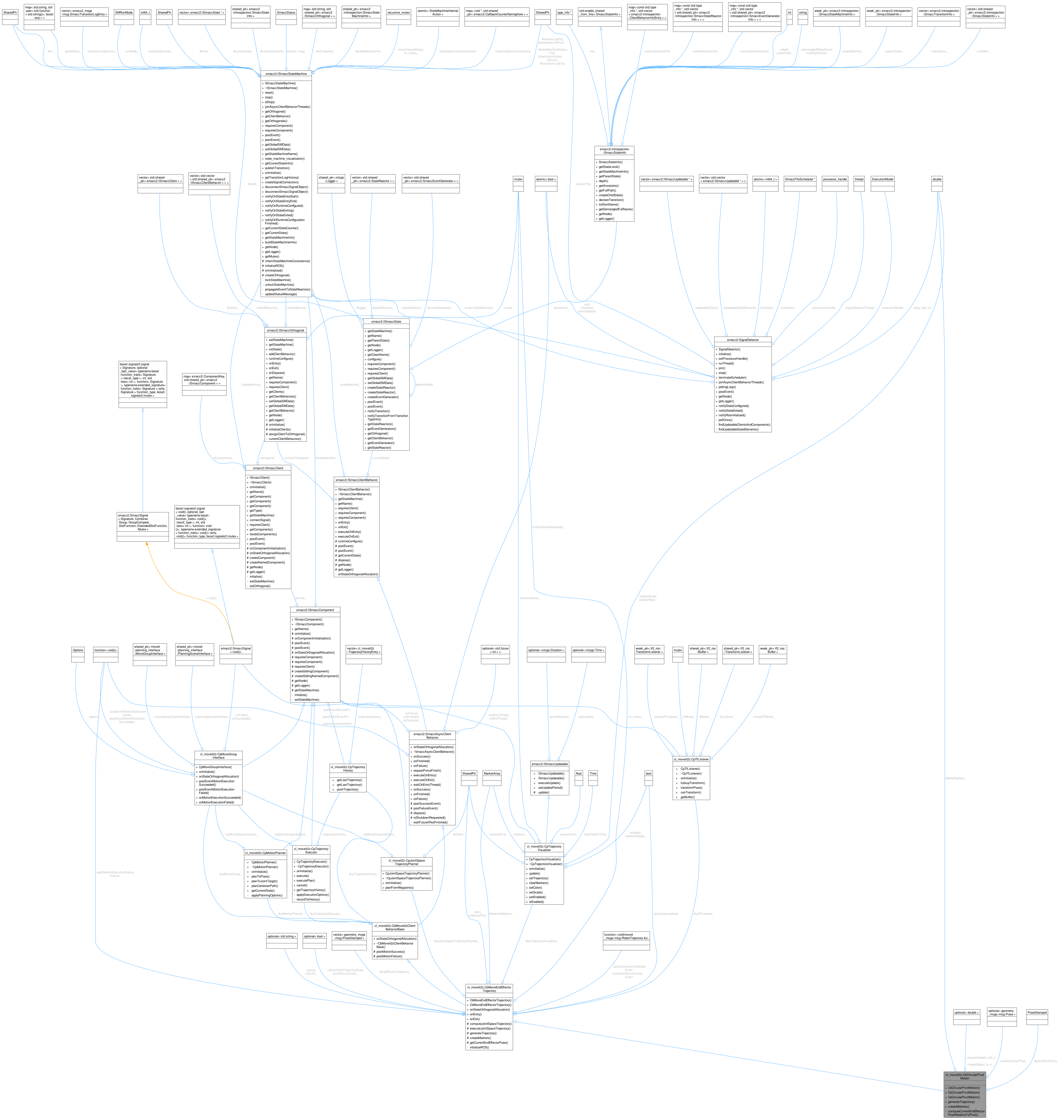
SharedPtr
map< std::string, std ::pair< std::function < std::string()>, boost ::any > >
vector< smacc2_msgs ::msg::SmaccTransitionLogEntry >
SMRunMode
int64_t
SharedPtr
vector< smacc2::ISmaccState * >
shared_ptr< smacc2 ::introspection::SmaccState Info >
SmaccStatus
map< std::string, std ::shared_ptr< smacc2 ::ISmaccOrthogonal > >
shared_ptr< smacc2 ::introspection::SmaccState MachineInfo >
recursive_mutex
atomic< StateMachineInternal Action >
map< void *, std::shared _ptr< smacc2::CallbackCounterSemaphore > >
SharedPtr
type_info *
std::enable_shared _from_this< SmaccStateInfo >
map< const std::type _info *, std::vector < smacc2::introspection ::ClientBehaviorInfoEntry > >
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccStateReactor Info > > >
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccEventGenerator Info > > >
int
string
weak_ptr< smacc2::introspection ::SmaccStateMachineInfo >
weak_ptr< smacc2::introspection ::SmaccStateInfo >
vector< smacc2::introspection ::SmaccTransitionInfo >
vector< std::shared _ptr< smacc2::introspection ::SmaccStateInfo > >
#nh_
-globalData_
-transitionLogHistory_
-runMode_
-stateSeqCounter_
#timer_
#currentState_
#currentStateInfo_
#status_msg_
#orthogonals_
#stateMachineInfo_
-eventQueueMutex_ -m_mutex_
-stateMachineCurrentAction
-stateCallbackConnections
#eventsLogPub_ #stateMachinePub_ #stateMachineStatusPub_ #transitionHistoryService_ #transitionLogPub_
+tid_
+staticBehaviorInfo
+stateReactorsInfo
+eventGeneratorsInfo
+depth_ +stateIndex_
+demangledStateName +fullStateName
+stateMachine_
+parentState_
+transitions_
+children_
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
smacc2::introspection ::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ getStateMachineInfo()
+ getParentState()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
-statusPub_
#node_
vector< std::shared _ptr< smacc2::ISmaccClient > >
vector< std::vector < std::shared_ptr< smacc2 ::ISmaccClientBehavior > > >
shared_ptr< rclcpp ::Logger >
vector< std::shared _ptr< smacc2::StateReactor > >
vector< std::shared _ptr< smacc2::SmaccEventGenerator > >
mutex
atomic< bool >
vector< smacc2::ISmaccUpdatable * >
vector< std::vector < smacc2::ISmaccUpdatable * > >
atomic< int64_t >
SmaccFifoScheduler *
processor_handle
thread
ExecutionModel
double
#clients_
-clientBehaviors_
-stateMachine_
#logger_
#stateReactors_
-signalDetector_
#eventGenerators_
-smaccStateMachine_
-mutex_
#stateInfo_
-end_ -initialized_ -rosInitialized_
-updatableClients_
-updatableStateElements_
-lastState_
-scheduler_
-processorHandle_
-signalDetectorThread_
-executionModel_
-loop_rate_hz
smacc2::ISmaccOrthogonal
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
- currentClientBehaviors()
smacc2::ISmaccState
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
#parentState_
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
boost::signals2::signal < Signature, optional _last_value< typename boost ::function_traits< Signature >::result_type >, int, std ::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost:: signals2::mutex >
map< smacc2::ComponentKey, std::shared_ptr< smacc2 ::ISmaccComponent > >
-stateMachine_
-stateMachine_
#components_
-orthogonal_
-currentOrthogonal
#stateMachine_
-currentState
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()
smacc2::SmaccSignal < Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
boost::signals2::signal < void(), optional_last _value< typename boost:: function_traits< void()> ::result_type >, int, std ::less< int >, function< void ()>, typename extended_signature < function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >
< void()>
-isShutdownRequested_
-markersMutex_
-markerScale_ -publishRate_
#owner_
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
Options
function< void()>
shared_ptr< moveit ::planning_interface ::MoveGroupInterface >
shared_ptr< moveit ::planning_interface ::PlanningSceneInterface >
smacc2::SmaccSignal < void()>
vector< cl_moveit2z ::TrajectoryHistoryEntry >
optional< std::future < int > >
optional< rclcpp::Duration >
optional< rclcpp::Time >
weak_ptr< tf2_ros:: TransformListener >
mutex
shared_ptr< tf2_ros ::Buffer >
shared_ptr< tf2_ros ::TransformListener >
weak_ptr< tf2_ros:: Buffer >
-options_
-postEventMotionExecution Failed_ -postEventMotionExecution Succeeded_
+moveGroupClientInterface
+planningSceneInterface
-onFailed_ -onSucceeded_
-postFailureEventFn
-postFinishEventFn
-postSuccessEventFn_
-trajectoryHistory_
-onFailure_ -onFinished_ -onSuccess_
-onEntryThread_ -onExitThread_
-periodDuration_
-lastUpdate_
-m_mutex_
-sharedTfListener_
-tfMutex_
-tfBuffer_
-tfListener_
-sharedTfBuffer_
cl_moveit2z::CpMoveGroup Interface
+ CpMoveGroupInterface()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ postEventMotionExecution Succeeded()
+ postEventMotionExecution Failed()
+ onMotionExecutionSucceeded()
+ onMotionExecutionFailed()
cl_moveit2z::CpTrajectory History
+ getLastTrajectory()
+ getLastTrajectory()
+ pushTrajectory()
smacc2::SmaccAsyncClient Behavior
+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()
SharedPtr
MarkerArray
smacc2::ISmaccUpdatable
+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ executeUpdate()
+ setUpdatePeriod()
# update()
float
Time
bool
cl_moveit2z::CpTfListener
+ CpTfListener()
+ ~CpTfListener()
+ onInitialize()
+ lookupTransform()
+ transformPose()
+ canTransform()
+ getBuffer()
-cpMoveGroupInterface_
-cpMoveGroupInterface_
-trajectoryHistory_
-cpMoveGroupInterface_
-ikClient_
-markersPub_
-markers_
-markerColor_
-lastPublishTime_
-enabled_ -markersReady_
cl_moveit2z::CpMotionPlanner
+ CpMotionPlanner()
+ ~CpMotionPlanner()
+ onInitialize()
+ planToPose()
+ planToJointTarget()
+ planCartesianPath()
+ getCurrentState()
- applyPlanningOptions()
cl_moveit2z::CpTrajectory Executor
+ CpTrajectoryExecutor()
+ ~CpTrajectoryExecutor()
+ onInitialize()
+ execute()
+ executePlan()
+ cancel()
+ getTrajectoryHistory()
- applyExecutionOptions()
- recordToHistory()
cl_moveit2z::CpJointSpace TrajectoryPlanner
+ CpJointSpaceTrajectoryPlanner()
+ ~CpJointSpaceTrajectoryPlanner()
+ onInitialize()
+ planFromWaypoints()
cl_moveit2z::CpTrajectory Visualizer
+ CpTrajectoryVisualizer()
+ ~CpTrajectoryVisualizer()
+ onInitialize()
+ update()
+ setTrajectory()
+ clearMarkers()
+ setColor()
+ setScale()
+ setEnabled()
+ isEnabled()
-postMotionExecutionFailure Events
#cpMoveGroup_
#cpTrajectoryHistory_
#cpMotionPlanner_
#cpTrajectoryExecutor_
-iksrv_ -markersPub_
#beahiorMarkers_
-autocleanmarkers
#cpTfListener_
#deltaRadians_
cl_moveit2z::CbMoveit2zClient BehaviorBase
+ onStateOrthogonalAllocation()
+ ~CbMoveit2zClientBehavior Base()
# postMotionSuccess()
# postMotionFailure()
optional< std::string >
optional< bool >
vector< geometry_msgs ::msg::PoseStamped >
function< void(moveit _msgs::msg::RobotTrajectory &)>
#cpJointSpaceTrajectoryPlanner_
#cpTrajectoryVisualizer_
+group_ +tipLink_
+allowInitialTrajectoryState JointDiscontinuity_
#endEffectorTrajectory_
-postIncorrectInitialState Event -postJointDiscontinuity Event
cl_moveit2z::CbMoveEndEffector Trajectory
+ CbMoveEndEffectorTrajectory()
+ CbMoveEndEffectorTrajectory()
+ onStateOrthogonalAllocation()
+ onEntry()
+ onExit()
# computeJointSpaceTrajectory()
# executeJointSpaceTrajectory()
# generateTrajectory()
# createMarkers()
# getCurrentEndEffectorPose()
- initializeROS()
optional< double >
optional< geometry _msgs::msg::Pose >
PoseStamped
+angularSpeed_rad_s _ +linearSpeed_m_s_
+relativeInitialPose_
#planePivotPose_
cl_moveit2z::CbCircularPivot Motion
+ CbCircularPivotMotion()
+ CbCircularPivotMotion()
+ CbCircularPivotMotion()
+ generateTrajectory()
+ createMarkers()
- computeCurrentEndEffector PoseRelativeToPivot()